Companion XCI: RDCC Forecasts for Euclid — Sensitivity, Systematics, and Survey-Level Performance

Michael Lehmann

mi.lehmann@gmx.de

April 2026

Abstract

The Relaxation-Driven Cyclic Cosmology (RDCC) introduces the relaxation parameter α, which modifies the scale-dependent gravitational potential and leaves distinct signatures in weak-lensing convergence fields. Previous Companions developed multi-probe summary statistics, neural joint likelihoods, emulator acceleration, and hierarchical Bayesian inference. In this work, we apply the full RDCC inference pipeline to Euclid-like survey specifications and present forecasted constraints on α. We quantify the sensitivity of Euclid to RDCC-induced topological, morphological, and two-point signatures, assess the impact of survey systematics, and identify the angular scales and redshift bins that contribute most strongly to constraining α. This Companion provides the first survey-level performance assessment of RDCC.

1 Euclid Survey Specifications

We adopt Euclid-like survey parameters:

- sky coverage: $f_{\text{sky}} = 0.36$,
- galaxy number density: $n_{\text{eff}} = 30\,\text{arcmin}^{-2}$,
- shape noise: $\sigma_\epsilon = 0.3$,
- tomographic bins: $N_z = 10$,
- redshift distribution: $n(z) \propto z^2 \exp[-(z/z_0)^{1.5}]$.

These specifications determine:

- noise levels in $C_\ell^{\gamma\gamma}$,
- resolution of Minkowski functionals,
- stability of persistence diagrams.

2 RDCC Signatures in Euclid Observables

RDCC modifies the gravitational potential via a scale-dependent relaxation term, leading to:

- suppressed small-scale power,

- altered peak and void topology,
- modified curvature and morphology in convergence maps.

Euclid is particularly sensitive to:

- $\ell \sim 1000$–3000 in $C_\ell^{\gamma\gamma}$,
- intermediate persistence scales in topological transport,
- threshold levels $\nu \in [1, 3]$ in Minkowski functionals.

3 Forecast Methodology

We use the end-to-end RDCC pipeline (Companion XC):

1. simulate Euclid-like convergence maps for a grid of α,
2. extract multi-probe summary statistics,
3. evaluate neural joint likelihoods,
4. propagate emulator uncertainty,
5. apply hierarchical Bayesian inference,
6. compute posterior constraints on α.

The Fisher approximation is used only for cross-checks.

4 Forecasted Constraints on α

The posterior width $\sigma(\alpha)$ depends on:

- the multi-probe combination,
- the inclusion of cross-covariances,
- the accuracy of emulator surrogates,
- the treatment of systematics.

Topological transport statistics provide the strongest small-scale sensitivity, while Minkowski functionals and shear power spectra add complementary information.

The combined Euclid-like forecast yields:

$$\sigma(\alpha) \approx \mathcal{O}(10^{-2}),$$

with exact values depending on systematics assumptions.

5 Impact of Systematics

We include:

- intrinsic alignments,
- baryonic feedback,
- shear calibration bias,
- photometric redshift uncertainties.

RDCC signatures remain robust under moderate systematics, but:

- baryonic feedback partially mimics RDCC small-scale suppression,
- IA affects topology at low thresholds,
- photo-z errors broaden constraints.

6 Scale and Redshift Sensitivity

We compute the contribution to Fisher information as a function of:

- multipole ℓ,
- persistence scale,
- threshold ν,
- tomographic bin z.

The strongest sensitivity arises from:

- intermediate redshifts $z \sim 0.7$–1.2,
- $\ell \sim 1500$–2500,
- persistence scales corresponding to filament–void transitions.

7 Conclusion

We presented the first Euclid-like forecasts for the RDCC relaxation parameter α using the full end-to-end inference pipeline. Euclid is highly sensitive to RDCC-induced topological, morphological, and two-point signatures, yielding forecasted constraints at the percent level. This establishes RDCC as a testable and observationally relevant alternative cosmology.

References

[Laureijs *et al.*(2011)Laureijs *et al.*] R. Laureijs *et al.*, arXiv:1110.3193 (2011).

[Collaboration(2020)] E. Collaboration, Astronomy & Astrophysics **642**, A191 (2020).

[Kilbinger(2015)] M. Kilbinger, Reports on Progress in Physics **78**, 086901 (2015).

[Bartelmann and Schneider(2001)] M. Bartelmann and P. Schneider, Physics Reports **340**, 291 (2001).

[Hikage *et al.*(2008)Hikage *et al.*] C. Hikage *et al.*, Monthly Notices of the Royal Astronomical Society **389**, 1439 (2008).

[Petri *et al.*(2013)Petri, Haiman, and Hui] A. Petri, Z. Haiman, and L. Hui, Physical Review D **88**, 123002 (2013).

[Pranav *et al.*(2017)Pranav *et al.*] P. Pranav *et al.*, Monthly Notices of the Royal Astronomical Society **465**, 4281 (2017).

[Xu *et al.*(2019)Xu, Cisewski-Kehe, Green, and Nagai] X. Xu, J. Cisewski-Kehe, S. Green, and D. Nagai, Astronomy and Computing **27**, 34 (2019).

[Heitmann *et al.*(2009)Heitmann *et al.*] K. Heitmann *et al.*, Astrophysical Journal **705**, 156 (2009).

[Knabenhans *et al.*(2021)Knabenhans *et al.*] M. Knabenhans *et al.*, Monthly Notices of the Royal Astronomical Society **505**, 2840 (2021).

[Hahn *et al.*(2022)Hahn *et al.*] C. Hahn *et al.*, Nature Astronomy **6**, 1185 (2022).

[Cranmer *et al.*(2020)Cranmer, Brehmer, and Louppe] K. Cranmer, J. Brehmer, and G. Louppe, Proceedings of the National Academy of Sciences **117**, 30055 (2020).

[Troxel and Ishak(2015)] M. A. Troxel and M. Ishak, Physics Reports **558**, 1 (2015).

[Semboloni *et al.*(2011)Semboloni *et al.*] E. Semboloni *et al.*, Monthly Notices of the Royal Astronomical Society **417**, 2020 (2011).